EduRAG Scholar

Official Client Project Handover & Production Specification Document

Client Name:	Asman Musa (makeromusa17@gmail.com)	**Date of Delivery:**	October 2026
Project Name:	EduRAG Scholar (Chiromo Academic Copilot)	**Deployment Status:**	**ACTIVE - Production Live & Verified**
Target Audience:	UoN Chiromo Campus - B.Sc. Computer Science	**Core Architecture:**	Multi-Tenant Hybrid RAG + Vision OCR

1. Executive Project Overview

EduRAG Scholar is an enterprise-grade academic learning assistant built for students taking the 7 core Year 2 Computer Science units at the University of Nairobi. It bridges lecture notes, past exam papers, and textbooks with autonomous AI retrieval, ensuring strict multi-user privacy, verified slide citations, and spaced repetition.

2. Production Deployment Endpoints

Resource	Provider / Host	Live Production URL
Frontend Web App	Vercel	https://learningassistant-chi.vercel.app
Backend API & Engine	Microsoft Azure	https://edurag-openai-musa-cpewa9epdmagg8b4.uaenorth-01.azurewebsites.net
Interactive Swagger Docs	Azure Web App	https://...azurewebsites.net/docs
Source Code Repository	GitHub	https://github.com/makeromusa/edurag-scholar
Database & Auth Engine	Supabase Cloud	https://xccrkalsujefsghtuirp.supabase.co
Document NoSQL Store	MongoDB Atlas	cluster0.ve6rbxy.mongodb.net (db: edurag)

3. Verified Deliverables & Completed Features

- **Strict Multi-Tenant Account Isolation:** Users authenticate via Google OAuth or verified Email/Password. Row-level filters ensure a user only queries and views materials they have personally uploaded. Classmates cannot access each other's data.
- **15+ MB Document & Textbook Ingestion:** High-volume PDF ingestion engine tested on 600+ page textbooks and 15+ MB materials. Includes automated Azure OpenAI gpt-4.1-mini Vision OCR fallback for image-only scans.
- **Interactive Slide & Citation Viewer:** AI answers include clickable [Slide #] citations that immediately open and highlight the corresponding slide page in the synchronized reader.
- **Responsive Mobile Architecture:** Slide-over drawer navigation and a mobile workstation switcher ([AI Copilot] | [Course Materials]) preventing layout breakage on iOS and Android phones.
- **Single Polished Sign-In Entry:** Streamlined single 'Sign In / Register' button in the top header, eliminating redundant buttons from the sidebar.
- **Spaced Repetition Suite (SM-2):** Interactive flashcard review system utilizing the SuperMemo SM-2 algorithm to schedule retention intervals based on difficulty scores.

- 3 AM Nightly Abstractor: Automated Chiromo lecture timetable synthesis and weakness drill delivery via background cron and SendGrid integration.

4. Environment & Security Configuration Schema

Notice: Sensitive production keys, database passwords, and secrets are securely masked below and stored exclusively in your local protected .env and cloud deployment settings (Vercel & Azure App Settings).

Variable Name	Component	Configured Value / Status
`SUPABASE_URL`	Supabase	https://xccrkalsujefsghtuirp.supabase.co
`SUPABASE_KEY`	Supabase	**[CONFIGURED SECURELY IN VAULT]**
`SUPABASE_SERVICE_ROLE_KEY`	Supabase Admin	**[CONFIGURED SECURELY IN VAULT]**
`MONGODB_URI`	MongoDB Atlas	**[PROTECTED ATLAS CLUSTER URI]**
`AZURE_OPENAI_ENDPOINT`	Azure AI	https://makeromusa17-0943-resource.../openai/v1
`AZURE_OPENAI_API_KEY`	Azure AI	**[CONFIGURED SECURELY IN VAULT]**
`AZURE_OPENAI_CHAT_DEPLOYMENT`	Azure AI	gpt-4.1-mini
`AZURE_OPENAI_EMBEDDING_DEPLOYMENT`	Azure AI	text-embedding-3-small
`TAVILY_API_KEY`	Search Fallback	**[CONFIGURED SECURELY IN VAULT]**
`SENDGRID_API_KEY`	SendGrid Email	**[CONFIGURED SECURELY IN VAULT]**
`SENTRY_DSN`	Observability	[CONFIGURED SENTRY INGESTION DSN]

5. Operations & Developer Maintenance Quickstart

Local Startup Commands:
- **Backend:** cd backend && venv\Scripts\activate && uvicorn app.main:app --reload (Port 8000)
- **Frontend:** cd frontend && npm run dev (Port 3000)

Automated Deployment:
- Pushes to the main GitHub branch trigger instant Vercel frontend rebuilds and Azure deployment workflows.

Client Sign-Off: Delivered complete, production-ready, and verified for Asman Musa.